



Prof Paul Rheeder
MMed (Int Med), FCP, PhD
(Clin Epi)

Dept Internal
Medicine

University of
Pretoria



Shiny Apps for Biostats

- 1. Why we need these Apps
- 2. Where to find and install these Apps
- 3. How to use these Apps



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Challenges

Note the names in the first row. Not useful in R. Note the mix of numbers and text in some columns

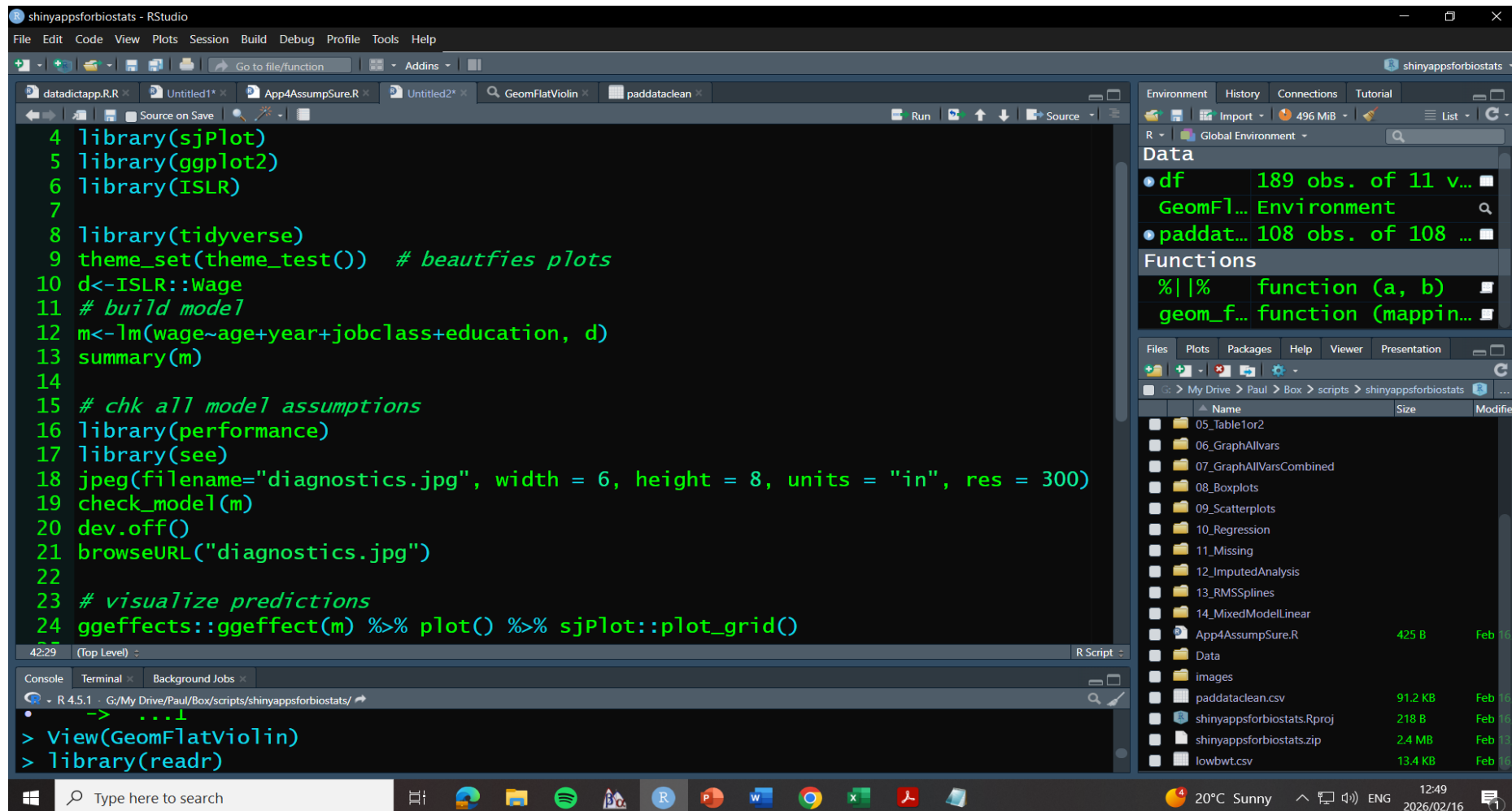
- Seems like no one knows how to make a valid Spreadsheet

Case	EF %	LAVI ml/ m²	E/e ratio	TR velocity	QTc m/s	Systolic dysfunction	Diastolic dysfunction	CTP score	MELD-na score	Age Years	Gender	BMI Kg/ m²
1-	65	21.6 ml/ m²	4.5	1.3m/s	409	No	No	B	17	32	Male	21.45
2-	77	11.9ml/ m²	6.7	1.3m/s	379	No	No	B	16	57	Female	19.12
3-	65	21.9 ml/ m²	5.6	1.4m/s	384	No	No	B	14	33	Female	25.11
4-	58	19.9 ml/ m²	5.9	1.8m/s	390	No	No	B	13	62	Female	28.21
5-	69	35 ml/ m²	7.9	1.89m/s	385	Yes	No	B	19	73	Female	27.16
6-	55	23ml/ m²	6.65	1.87m/s	379	No	No	C	22	43	Male	23.02
7-	66	23ml/ m²	5.3	1.97m/s	401	No	No	B	19	55	Male	21.24
8-	56	29ml/ m²	4.6	1.28m/s	394	No	No	C	24	45	Female	31.23
9-	57	28ml/ m²	7.2	1.6m/s	405	No	No	B	18	39	Male	23.56
10-	55	18.3ml/m²	5.81	1.47m/s	434	No	No	B	17	56	Female	21.42
11-	66	11.8ml/ m²	3.12	1.32m/s	410	No	no	C	23	63	Male	28.54
12-	63	25.4	6.68	1.08m/s	413	No	No	C	22	68	Female	22.61
13-	57	35ml/ m²	5.12	2.3m/s	378	Yes	No	C	23	33	Male	21.74
14-	63	24ml/ m²	5.12	1.69m/s	402	No	No	C	30	46	Female	34.71
15-	73	28ml/ m²	2.4	1.69m/s	388	No	No	B	19	22	Female	24.34
16-	46	19ml/ m²	4.37	1.5m/s	419	Yes	No	C	26	37	Male	21.77
17-	63	24ml/ m²	2.6	1.78m/s	412	No	No	C	21	66	Female	21.66
18-	73	18ml/ m²	4.68	1.5m/s	451	No	No	B	19	56	Male	20.74
19-	69	22ml/ m²	4.97	1.5m/s	443	No	No	C	23	45	Female	19.74
20-	49	36m/ m²	4.95	1.5m/s	405	Yes	Yes	C	27	51	Male	30.17
21-	45	39.5ml/ m²	6.74	1.81m/s	403	Yes	Yes	C	30	69	Female	28.76
22-	57	44.7ml/ m²	3.55	2.91m/s	386	No	Yes	C	24	78	Male	31.08
23-	65	21ml/ m²	4.56	2.2m/s	429	No	Yes	C	22	42	Female	22.67

Challenges

- Everything requires coding!

R has a steep learning curve. Lot of coding needed!



The screenshot displays the RStudio environment with the following components:

- Source Editor:** Contains R code for loading libraries, setting themes, fitting a linear model, checking assumptions, and visualizing results.
- Environment:** Shows loaded data frames and functions.
- Files:** Lists project files and folders.
- Console:** Shows the execution of the code.

```
4 library(sjPlot)
5 library(ggplot2)
6 library(ISLR)
7
8 library(tidyverse)
9 theme_set(theme_test()) # beautifies plots
10 d<-ISLR::Wage
11 # build model
12 m<-lm(wage~age+year+jobclass+education, d)
13 summary(m)
14
15 # chk all model assumptions
16 library(performance)
17 library(se)
18 jpeg(filename="diagnostics.jpg", width = 6, height = 8, units = "in", res = 300)
19 check_model(m)
20 dev.off()
21 browseURL("diagnostics.jpg")
22
23 # visualize predictions
24 ggeffects::ggeffect(m) %>% plot() %>% sjPlot::plot_grid()
```

Environment:

- Data:**
 - df: 189 obs. of 11 v...
 - GeomFl... Environment
 - paddat... 108 obs. of 108 ...
- Functions:**
 - %|>% function(a, b)
 - geom_f... function(mappin...

Files:

Name	Size	Modified
05_Table1or2		
06_GraphAllVars		
07_GraphAllVarsCombined		
08_Boxplots		
09_Scatterplots		
10_Regression		
11_Missing		
12_ImputedAnalysis		
13_RMSSplines		
14_MixedModelLinear		
App4AssumpSure.R	425 B	Feb 16
Data		
images		
paddataclean.csv	91.2 KB	Feb 16
shinyappsforbiostats.Rproj	218 B	Feb 16
shinyappsforbiostats.zip	2.4 MB	Feb 13
lowbwt.csv	13.4 KB	Feb 16

Console:

```
> view(GeomFlatViolin)
> library(readr)
```


Challenges

you want output to be seen in Word easily, e.g regression tables

- Getting results into publishable tables

The screenshot shows the RStudio interface with a script editor at the top containing two lines of code: `1 # works great with compile report in html` and `2`. The console at the bottom displays the output of a linear regression model. The residuals are shown as a table with columns Min, 1Q, Median, 3Q, and Max. The coefficients are shown as a table with columns Estimate, Std. Error, t value, and Pr(>|t|). The significance codes are shown as a string: `0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1`. The residual standard error is 35.83 on 2992 degrees of freedom. The multiple R-squared is 0.2646 and the adjusted R-squared is 0.2629. The F-statistic is 153.8 on 7 and 2992 DF, with a p-value of < 2.2e-16.

```
Residuals:
      Min       1Q   Median       3Q      Max
-109.944  -19.749   -3.715   14.613   216.209

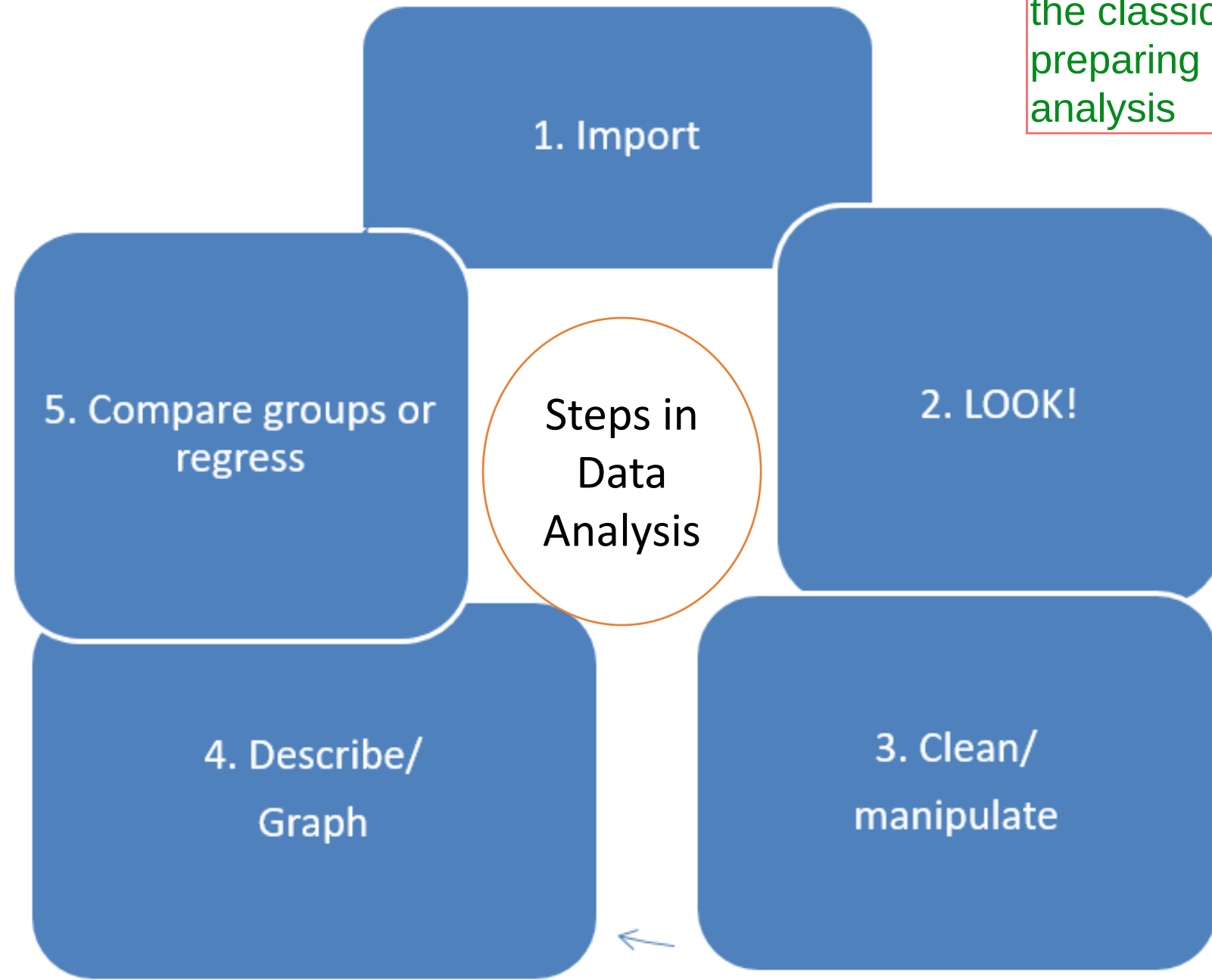
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -2.089e+03  6.483e+02  -3.222 0.001286 **
age          5.485e-01  5.719e-02   9.591 < 2e-16 ***
year         1.071e+00  3.233e-01   3.314 0.000931 ***
jobclass2. Information 4.575e+00  1.379e+00   3.318 0.000917 ***
education2. HS Grad   1.116e+01  2.473e+00   4.511 6.71e-06 ***
education3. Some College 2.339e+01  2.614e+00   8.947 < 2e-16 ***
education4. College Grad 3.834e+01  2.616e+00  14.655 < 2e-16 ***
education5. Advanced Degree 6.275e+01  2.871e+00  21.859 < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 35.83 on 2992 degrees of freedom
Multiple R-squared:  0.2646,    Adjusted R-squared:  0.2629
F-statistic: 153.8 on 7 and 2992 DF,  p-value: < 2.2e-16
```

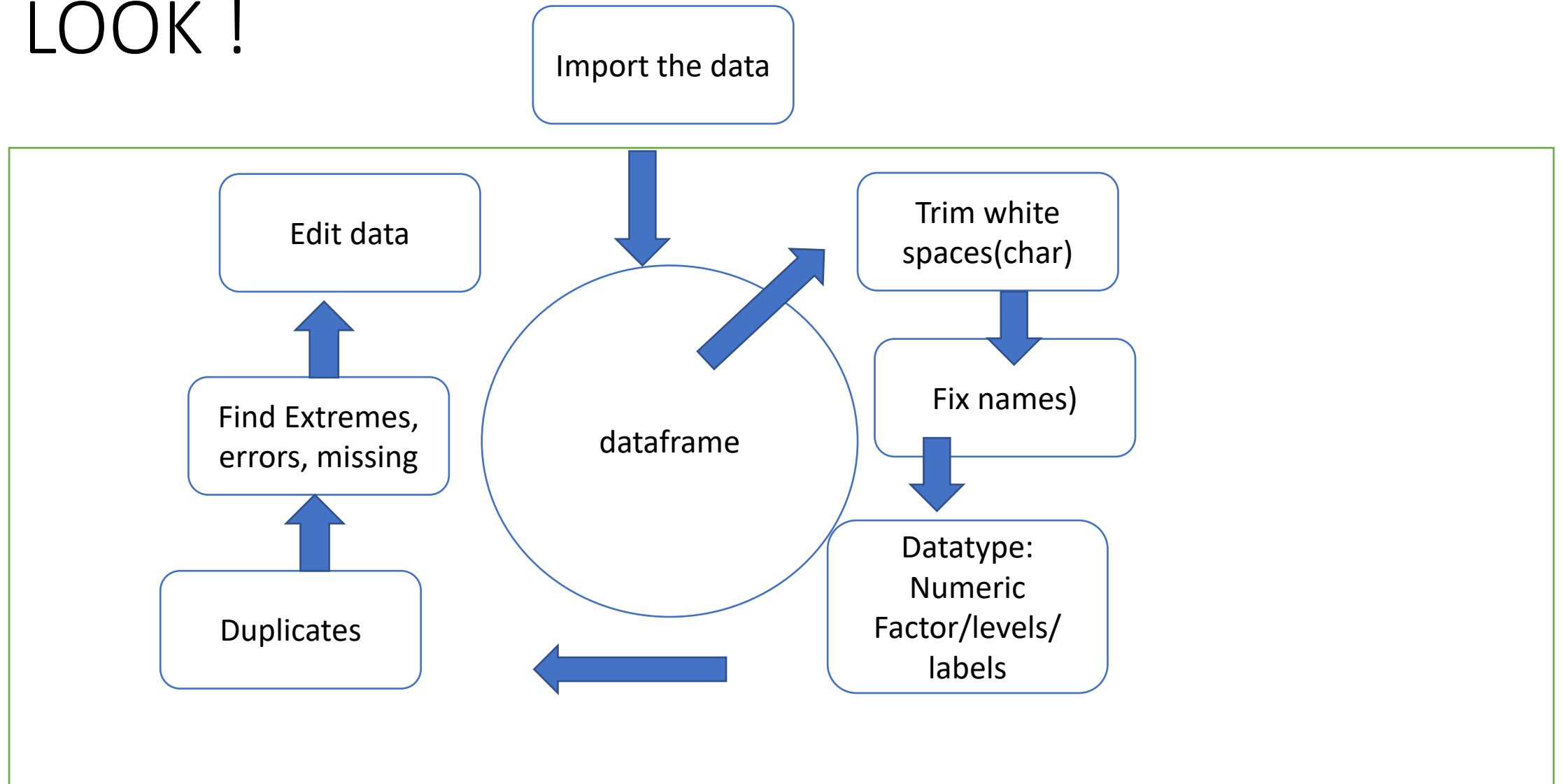
Challenges

- Students do not understand the stats you did!
- They should do their own descriptive stats and graphs
- They should do basic regression and interpret output

the classical steps of
preparing data before
analysis



LOOK !



USING SHINY APPS FOR BIOSTATISTICS

Here we have an App for each of these steps.

- 
1. DataDictionary: merge data with a dictionary file
 2. PostImport: open file and fix data format and names, choose data types
 3. DplyrEdit: do data manipulation with Dplyr
 4. Table1AllStats: look at/explore the data
 5. Table1or2: make nice tables, single or more groups
 6. GraphAllvars: graph the numerical and categorical variables
 7. GraphAllVarsComined: graph numerical and categorical variables, basic box and scatterplots
 8. Boxplots
 9. Scatterplots
 10. Regression: linear and logistic regression, test assumptions, show detail such as goodness of fit and calibration, show plots.

Here we have an App for each and every step including descriptive stats, comparing groups as well as regression

<https://github.com/paulrheeder-rgb/shinyappsforbiostats>

Demo of Apps



paul.rheeder@up.ac.za

- 01_DataDictionary
- 02_PostImport
- 03_DplyrDataEdit
- 04_Table1AllStats
- 05_Table1or2
- 06_GraphAllvars
- 07_GraphAllVarsCombined
- 08_Boxplots
- 09_Scatterplots
- 10_Regression
- 11_Missing
- 12_ImputedAnalysis
- 13_RMSSplines
- 14_MixedModelLinear

these Apps can be found on github.
mail me with comments or suggestions!

Windows taskbar showing icons for Google Chrome, Microsoft Word, and other applications. The system tray shows the date and time: 06:42 Thursday 2026/03/19.

Browser tabs: Google Gemini, Settings - Downloads. Address bar: chrome://settings/downloads. Bookmarks: Discover, Harrison's Principles..., 1 Defining counterf..., Course: Introductio..., An Introduction to '...'. Profile: School.

Settings sidebar menu:

- Settings
- Privacy and security
- Performance
- Appearance
- Search engine
- Default browser
- On startup
- Languages
- Downloads**
- Accessibility
- System
- Reset settings

Search settings

Your [profile is managed](#) by up.ac.za

Downloads settings card:

- Location: C:\Users\U02552183\Downloads [Change](#)
- Ask where to save each file before downloading: ☒
- Show downloads when they're done: ☒

In your browser make sure you select these options. This will allow you to save the output of each app in a folder of your choice.

Windows taskbar showing icons for File Explorer, Edge, PowerPoint, and a notification area with Desktop, ENG, and date/time (18:27 Friday 2026/03/13).

Browser tabs and address bar showing the GitHub repository: `github.com/paulrheeder-rgb/shinyappsforbiostats`. The address bar includes navigation buttons, search, and a user profile labeled "School".

README



In github make sure you read the instructions found in the README file

USING SHINY APPS FOR BIOSTATISTICS

Data analysis always has several steps: import data, fix format and clean names and white spaces, fix data types (such as factors with levels), check missing values, check and manage extreme values, do exploratory graphs and descriptive statistics, create tables, more box and scatter plots and finally regression analysis.

Here we have an App for each of these steps.

1. DataDictionary: merge data with a dictionary file
2. PostImport: open file and fix data format and names, choose data types
3. DplyrEdit: do data manipulation with Dplyr
4. Table1AllStats: look at/explore the data
5. Table1or2: make nice tables, single or more groups
6. GraphAllvars: graph the numerical and categorical variables
7. GraphAllVarsComined: graph numerical and categorical variables, basic box and scatterplots
8. Boxplots
9. Scatterplots
10. Regression: linear and logistic regression, test assumptions, show detail such as goodness of fit and calibration, show plots.

How to get started:

STEP 1: Install R <https://cran.r-project.org>

Get started

STEP 1: Install R <https://cran.r-project.org>

STEP 2: Install RStudio

<https://posit.co/download/rstudio-desktop/>

STEP 3: Find zipped folder

Follow these steps exactly:

Open this link in your web browser 

<https://github.com/paulrheeder-rgb>

click on shinyappsforbiostats

On the GitHub page, click the green "Code" button (top right)

Click "Download ZIP"

Make sure you have R and RStudio.
Download the zip file.
Create a folder called shinyappsforbiostats
In RStudio go to "open project" and click in the project file in this folder.
All the Apps should be seen in the right lower pane.

 Go to file

 Code ▾

About

Go to your Downloads folder

Right-click the ZIP file → choose "Extract All..." → click Extract

Extract to Documents on your PC (or any other folder) Not to Documents/shinyapps..... just Documents/

or in a drive eg google drive so you can access anywhere

You should now have:

Documents/ └─ shinyappsforbiostats-main/

STEP 4: Rename the folder (recommended)

Right-click the folder:

shinyappsforbiostats-main

Rename it to:

shinyappsforbiostats

STEP 5: Open RStudio

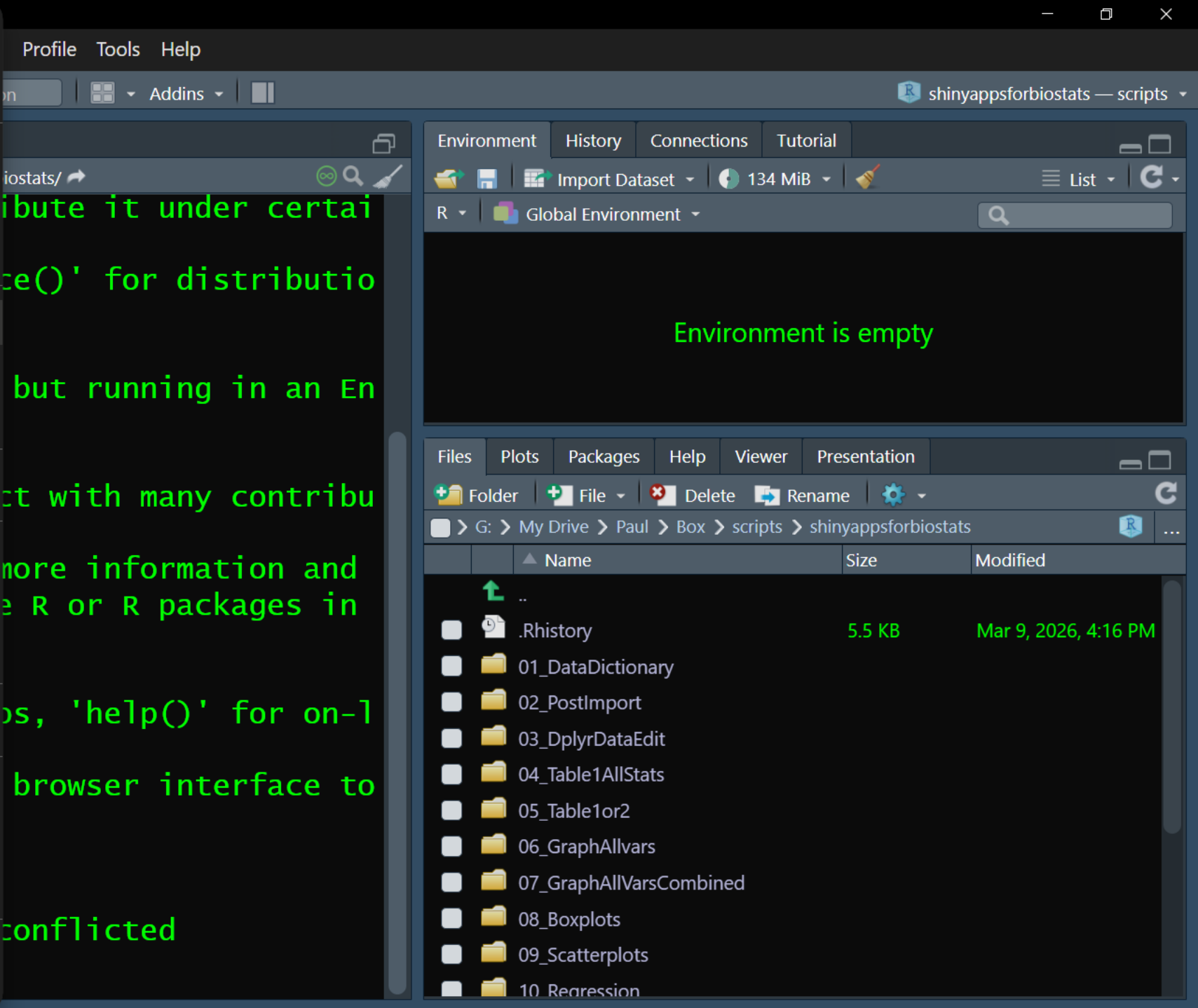
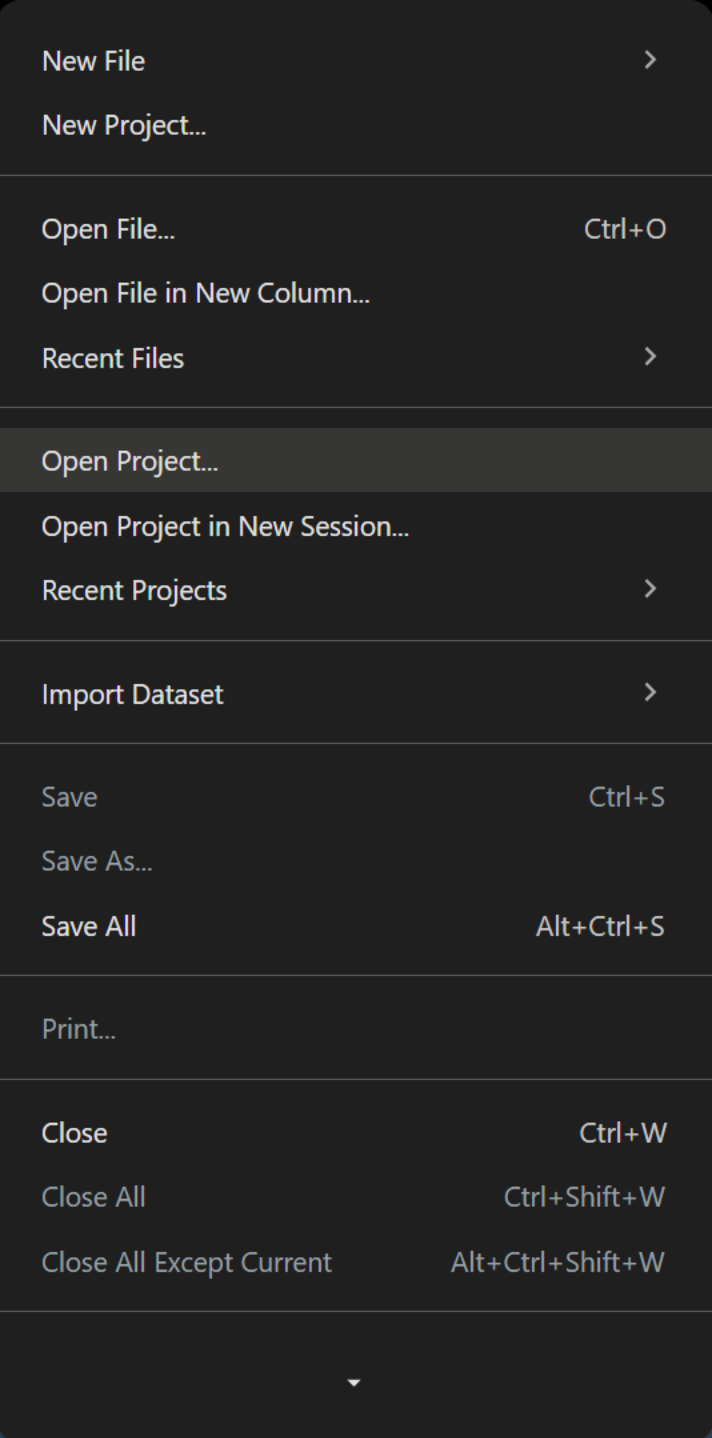
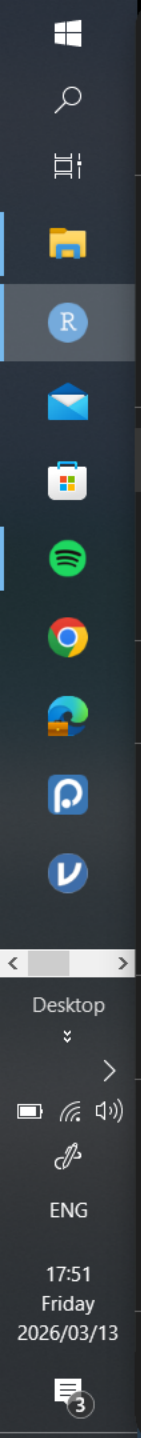
choose file>>>Open Project>>>> go to the shinyappsforbiostats folder click on the shinyappsfor biostats project file

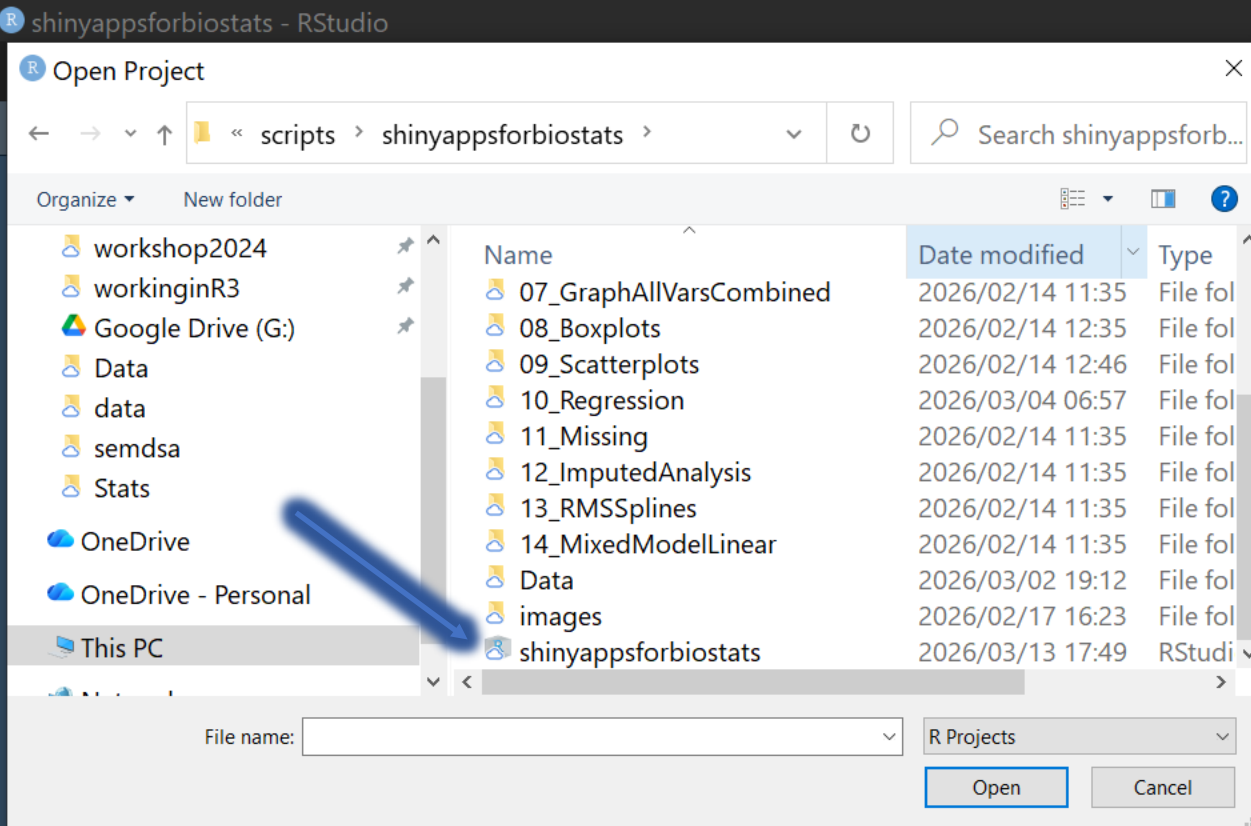
This project will now open in RStudio and in the Right pane under files you will see all the Apps

STEP 6: Open an app

Open the folder and click on app.R The is app will appear as a script file At the top of the file it will say which libraries you need to install

Step 6: Then choose run App





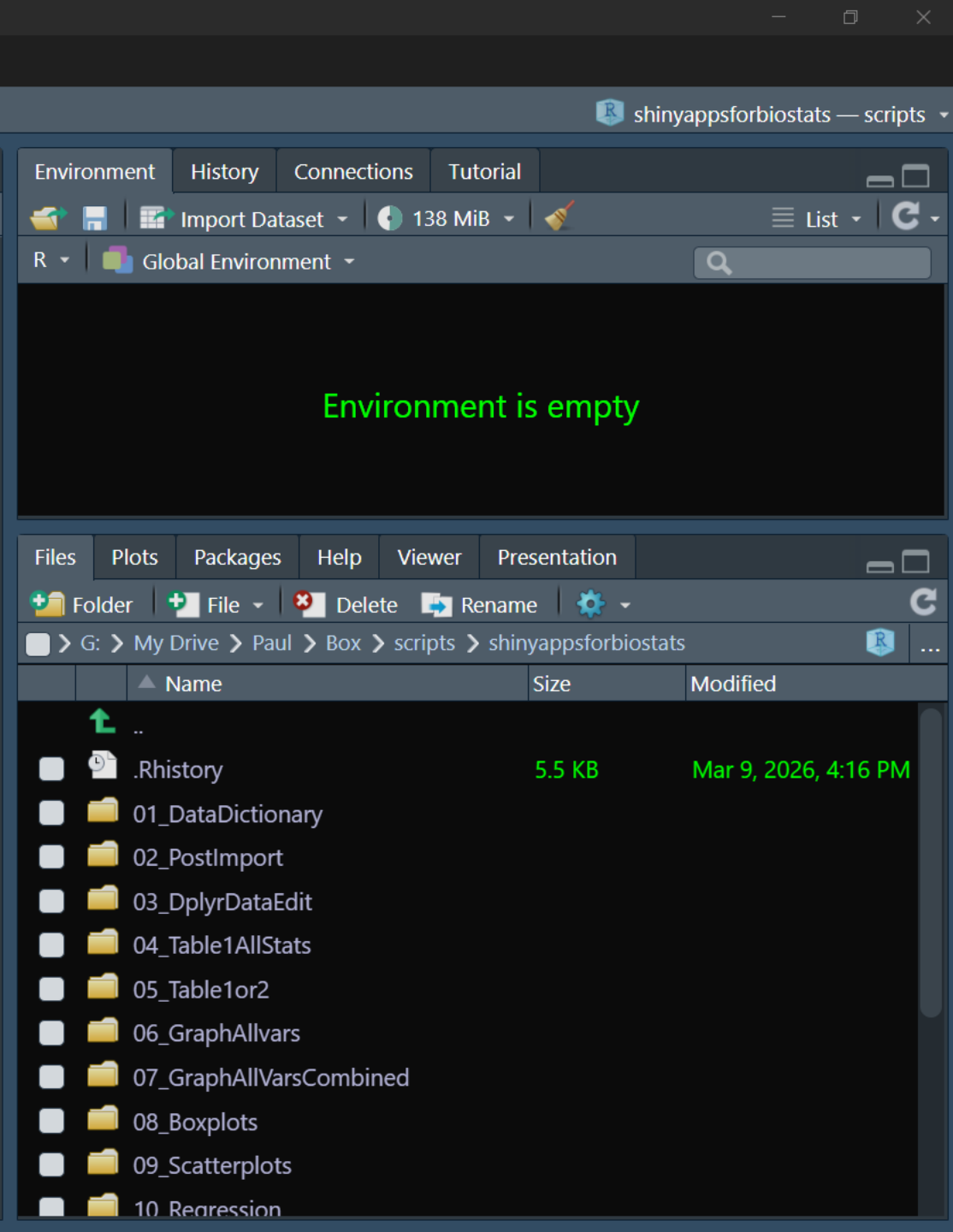
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.

Type 'q()' to quit R.

Loading required package: conflicted

> █



Windows taskbar icons: Start button, Search, Task View, File Explorer, RStudio, Mail, Calendar, Spotify, Chrome, Edge, Firefox, and system tray icons (Network, Volume, ENG, 17:53 Friday 2026/03/13, 3 notifications).

shinyappsforbiostats - RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Console Terminal Background Jobs

R 4.5.2 · G:/My Drive/Paul/Box/scripts/shinyappsforbiostats/

```
You are welcome to redistribute it under certain conditions.  
Type 'license()' or 'licence()' for distribution details.  
  
Natural language support but running in an English locale  
  
R is a collaborative project with many contributors.  
Type 'contributors()' for more information and  
'citation()' on how to cite R or R packages in publications.  
  
Type 'demo()' for some demos, 'help()' for on-line help, or  
'help.start()' for an HTML browser interface to help.  
Type 'q()' to quit R.  
  
Loading required package: conflicted  
> |
```

shinyappsforbiostats — scripts

Environment History Connections Tutorial

Import Dataset 138 MiB List

R Global Environment

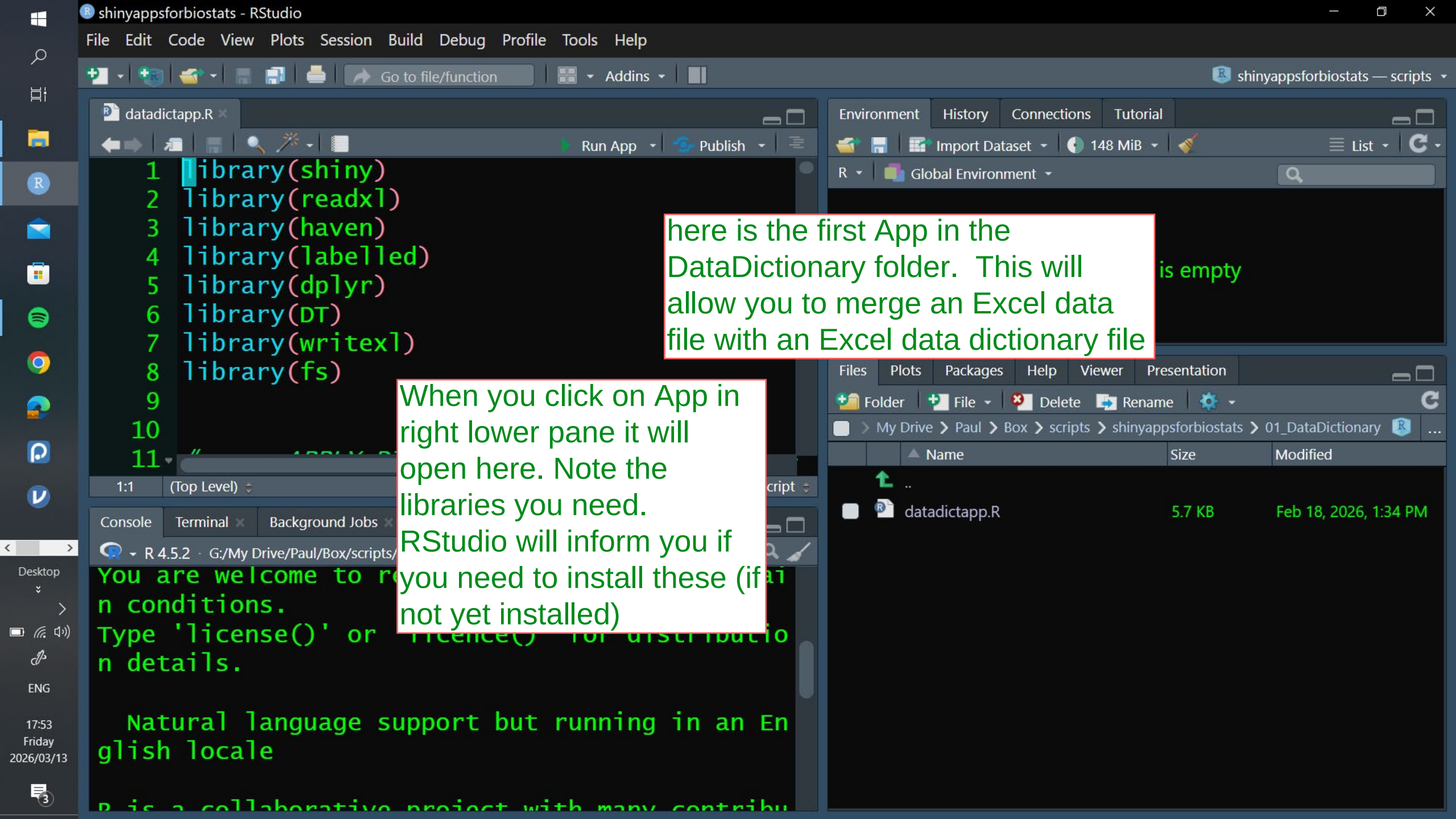
Environment is empty

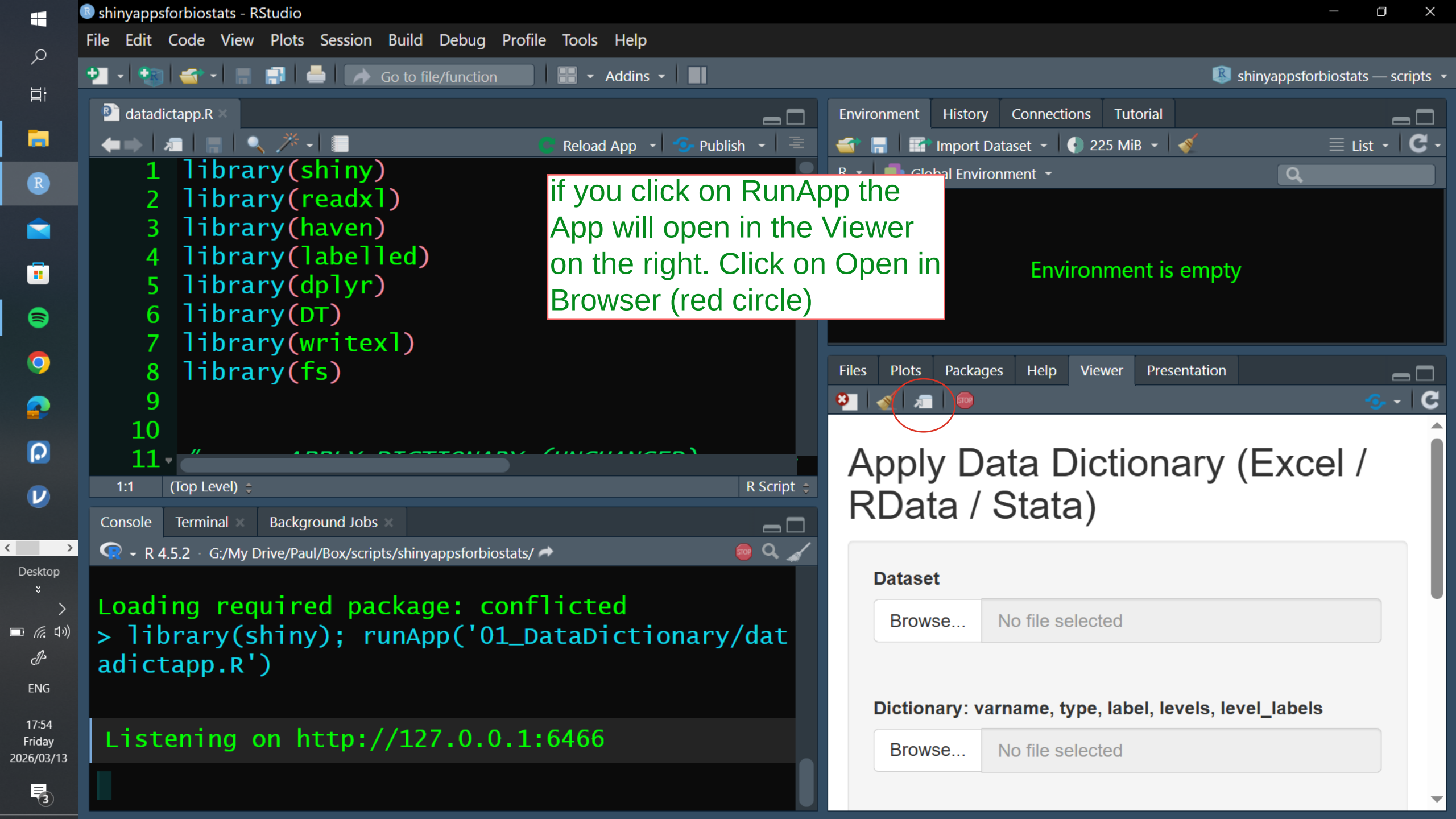
Files Plots Packages Help Viewer Presentation

Folder File Delete Rename

G: > My Drive > Paul > Box > scripts > shinyappsforbiostats

	Name	Size	Modified
..	..		
	.Rhistory	5.5 KB	Mar 9, 2026, 4:16 PM
	01_DataDictionary		
	02_PostImport		
	03_DplyrDataEdit		
	04_Table1AllStats		
	05_Table1or2		
	06_GraphAllvars		
	07_GraphAllVarsCombined		
	08_Boxplots		
	09_Scatterplots		
	10_Regression		





Apply Data Dictionary (Excel / RData / Stata)

01_DataDictionary

Dataset

Browse... cleaned_data.xlsx

Upload complete

Dictionary: varname, type, label, levels, level_labels

Browse... datadict_cleaned.xlsx

Upload complete

Apply Dictionary

Base name (no extension)

cleaned_data

Download cleaned data (.RData)

Download dictionary (.xlsx)

Cleaned data preview

Show 10 entries

Search:

Here you have the App open with data and dictionary loaded. Just Apply dictionary and save this file which will now have the right names and data types

	id	trgl	ldl	creat	tbir	tbi
	3	102	0.7786884903907776	0.7377048730850		
	000190734863	58	1.120301008224487	0.849623978131		
	999952316284	66	0.7756410241127014	0.80128198862075		
	4	4	1.799999952316284	5.400000095367432	59	1.133332967758179
	5	5	0.5	1.700000047683716	122	0.9391891956329346
	6	6	1.299999952316284	2.099999904632568	95	1.114503979682922
	7	7	3.700000047683716	4	68	0.8383234143257141

Data Loader & Variable Manager

02_PostImport

Load Dataset

Upload Excel / Stata / RData / RDS

Browse...

padR4.dta

Upload complete

Or select dataset from R environment:

Load Selected Dataset

Data Cleaning Options

Convert to data.frame

Trim whitespace (character columns)

Lowercase column names

Remove underscores in names

Dataset Preview

With the PostImport app you will load your data from Excel, Stata or R. Then convert to a dataframe, fix white spaces and names.

Search:

	at	tbir	tbil
102	0.7786884903907776	0.7377048730850	
2	2	1.399999976158142	4.800000190734863
3	3	0.8999999761581421	2.799999952316284
4	4	1.799999952316284	5.400000095367432
5	5	0.5	1.700000047683716

Showing 1 to 5 of 108 entries

Previous

1

2

3

4

5

...

22

Next

Variable Summary (complete unique values)

Show 20 entries

Search:

Variable

Class

N_unique

Unique_values

Trim whitespace (character columns)

Lowercase column names

Remove underscores in names

Duplicate Check

Show Duplicate Rows Remove Duplicates

Select variable(s)

id

Rename selected variable to:

study number

Apply Rename

Convert selected variable(s) to:

☐ Numeric

☐ Factor

☒ Character

Variable Summary (complete unique values)

Show 20 entries Search:

Variable	Class	N_unique	Unique_values
		108	
		37	
		39	
		61	
		105	
		103	
tbil	numeric	4	1, 2, 3, 4
race	numeric	2	0, 1
sex	numeric	101	
egfr	numeric	87	
b2m	numeric	54	
hba1c	numeric	3	0, 1, 2
smo	numeric		

You can show and remove duplicate rows

You can rename variables and mots important choos the datatype you want eg numeric or factor or character

Apply Rename

Convert selected variable(s) to:

- ☐ Numeric
- ☒ Factor
- ☐ Character

Factor levels (comma separated)

1, 2, 3, 4

Factor labels (comma separated)

Black, White, Indian, Other

Variable label (e.g. Sex of patient)

Ethnicity

Apply Conversion

Create New Factor with case_when

race	factor	4	White, Indian, Black, Other
------	--------	---	-----------------------------

Here you can see we chose race and made a factor with 4 levels

hba1c	numeric	54	
smo	numeric	3	0, 1, 2
dmttype	numeric	2	1, 2
hyp	numeric	2	0, 1
pad	numeric	2	0, 1
age	numeric	33	

Showing 1 to 16 of 16 entries

[Previous](#)

1

Next

Structure & Labels

```
'data.frame': 108 obs. of 16 variables:
 $ id      : num 1 2 3 4 5 6 7 8 9 10 ...
 ..- attr(*, "format.stata")= chr "%9.0g"
 $ trgl    : num 1.5 1.4 0.9 1.8 0.5 ...
```

Upload Excel, Stata, or RData

Browse...

padR4.dta

Upload complete

Or select dataset from R environment:

Load Selected Dataset

2. Data Operations

Choose Operation:

Rename Columns

Select Columns

Rename Columns

Filter Rows

Arrange Rows

Mutate (if_else)

Mutate (case_when)

Mutate (expression)

Recode Variable

Show 10 entries

03_DplyrDataEdit

Search:

	id	trgl	ldl	creat	tbir	tbi
1	1	1.5	3	102	0.7786884903907776	0.7377048730850
2	2	1.399999976158142	4.800000190734863	58	1.120301008224487	0.8496239781350
3	3	1.299999952316284	2.099999904632568	95	1.114503979682922	1.0458019971845
4	4	3.700000047683716	4	68	0.8383234143257141	0.850299417972500
5	5	1.5	2.200000047683716	95	0.8110235929489136	0.874015688896100
6	6	99	2.5	86	0.8364779949188232	0.924528300762100
7	7	2.400000095367432	2.700000047683716	56	0.783333420753479	0.550000011920000
8	8	0.5	1.700000047683716	122	0.9391891956329346	0.790540516376450
9	9	1.299999952316284	2.099999904632568	95	1.114503979682922	1.045801997184500
10	10	1.399999976158142	4.800000190734863	58	1.120301008224487	0.849623978135000

Showing 1 to 10 of 108 entries

Previous

1

2

3

4

5

...

11

Next

04_Table1AllStats

Upload Excel / Stata / RData / RDS / CSV

Browse... lowbwt.RData
Upload complete

Or select dataset from R environment:

[Dropdown menu]
Load Selected Dataset

Data Subset Options

Subset Mode:

Use all data
Apply Subset

Summary Table Options

☒ Include Categorical Variables

Group By (optional):

None

No grouping applied

Descriptive Summary

Characteristic

N = 189¹

id	121.08 (63.30) [4.00, 226.00] 123.00 [68.00, 176.00]
low	
>= 2500 g	130 (69%)
< 2500 g	59 (31%)
age	23.24 (5.30) [14.00, 45.00] 23.00 [19.00, 26.00]
lwt	129.81 (30.58) [80.00, 250.00] 121.00 [110.00, 140.00]

The Table1AllStats App allows you to look at your data with n(%) mean sd min max and median IQR

Black	26 (14%)
Other	67 (35%)
smoke	
No	115 (61%)

Upload Excel / Stata / RData / RDS / CSV

Browse... lowbwt.RData

Upload complete

Or select dataset from R environment:

Load Selected Dataset

Data Subset Options

Subset Mode:

Select columns

Select columns to keep:

- ☐ id
- ☒ low
- ☒ age
- ☒ lwt
- ☒ race
- ☒ smoke
- ☐ ptl

Grouping by: low

	Descriptive Summary	
	>= 2500 g N = 130 ¹	< 2500 g N = 59 ¹
age	23.66 (5.58) [14.00, 45.00] 23.00 [19.00, 28.00]	22.31 (4.51) [14.00, 34.00] 22.00 [19.00, 25.00]
lwt	133.30 (31.72) [85.00, 250.00] 123.50 [113.00, 147.00]	122.14 (26.56) [80.00, 200.00] 120.00 [103.00, 130.00]
race		
White	73 (56%)	23 (39%)
Black	15 (12%)	11 (19%)
No	86 (66%)	29 (49%)
Yes	44 (34%)	30 (51%)

¹ Mean (SD) [Min, Max] Median [Q1, Q3]; n (%)

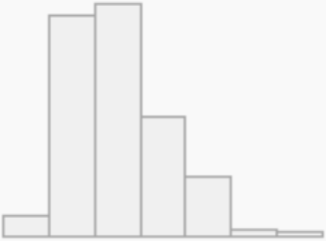
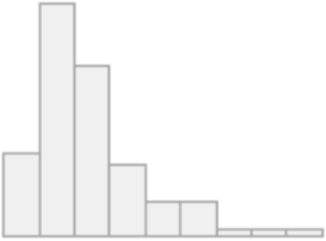
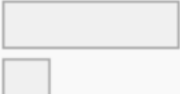
You can do it with one or more groups.
This is just to explore
No hypothesis tests

Data Frame Summary

Dimensions: 189 x 5

Duplicates: 10

You can use all variables or select variables and get a data summary

No	Variable	Stats / Values	Freqs (% of Valid)	Graph	Valid	Missing
1	low [factor]	1. ≥ 2500 g 2. < 2500 g	130 (68.8%) 59 (31.2%)		189 (100.0%)	0 (0.0%)
2	age [integer]	Mean (sd) : 23.2 (5.3) min $\leq$ med $\leq$ max: 14 $\leq$ 23 $\leq$ 45 IQR (CV) : 7 (0.2)	24 distinct values		189 (100.0%)	0 (0.0%)
3	lwt [integer]	Mean (sd) : 129.8 (30.6) min $\leq$ med $\leq$ max: 80 $\leq$ 121 $\leq$ 250 IQR (CV) : 30 (0.2)	75 distinct values		189 (100.0%)	0 (0.0%)
4	race	1. White 2. Black	96 (50.8%) 26 (13.8%)		189	0

Upload Excel/Stata/RData/CSV

Browse...

lowbwt.RData

Upload complete

Or select dataset from R environment:

Load Selected Dataset

Grouping variable

None

Parametric (mean $\pm$ SD)

age lwt bwt

Non-parametric (median [p25, p75])

Categorical (N, %)

race smoke

05_Table1or2

This App allows you to get a publication ready Table 1 (one group) or Table 2 (more than 1 group)

Descriptive statistics (overall)	
Variable	N = 189 ¹
age	23.2 (5.3)
lwt	129.8 (30.6)
bwt	2,944.7 (729.0)
race	
White	96 (51%)
Black	26 (14%)
Other	67 (35%)
smoke	
No	115 (61%)
Yes	74 (39%)

¹ Mean (SD); n (%)

Upload Excel/Stata/RData/CSV

Browse... lowbwt.RData
Upload complete

Or select dataset from R environment:

[Dropdown menu]
Load Selected Dataset

Grouping variable

low

Parametric (mean ± SD)

age lwt bwt

Non-parametric (median [p25, p75])

[Empty input field]

Categorical (N, %)

race smoke

You choose what you want: parametric, non parametric or categories

Descriptive statistics by group

Variable	≥ 2500 g N = 130 ¹	< 2500 g N = 59 ¹	p-value ²
age	23.7 (5.6)	22.3 (4.5)	0.078
lwt	133.3 (31.7)	122.1 (26.6)	0.013
bwt	3,329.3 (477.7)	2,097.1 (390.9)	<0.001
race			0.082
White	73 (56%)	23 (39%)	
Black	15 (12%)	11 (19%)	
Other	42 (32%)	25 (42%)	
smoke			0.040
No	86 (66%)	29 (49%)	
Yes	44 (34%)	30 (51%)	

¹ Mean (SD); n (%)
² Welch Two Sample t-test; Chi-squared test

Select data source:

☒ Upload file ☐ R Environment

Upload Excel or Stata or RData

Browse...

lowbwt.RData

Upload complete

Select Numeric Variables:

age

Select Factor Variables:

low race

Categorical Plot Type:

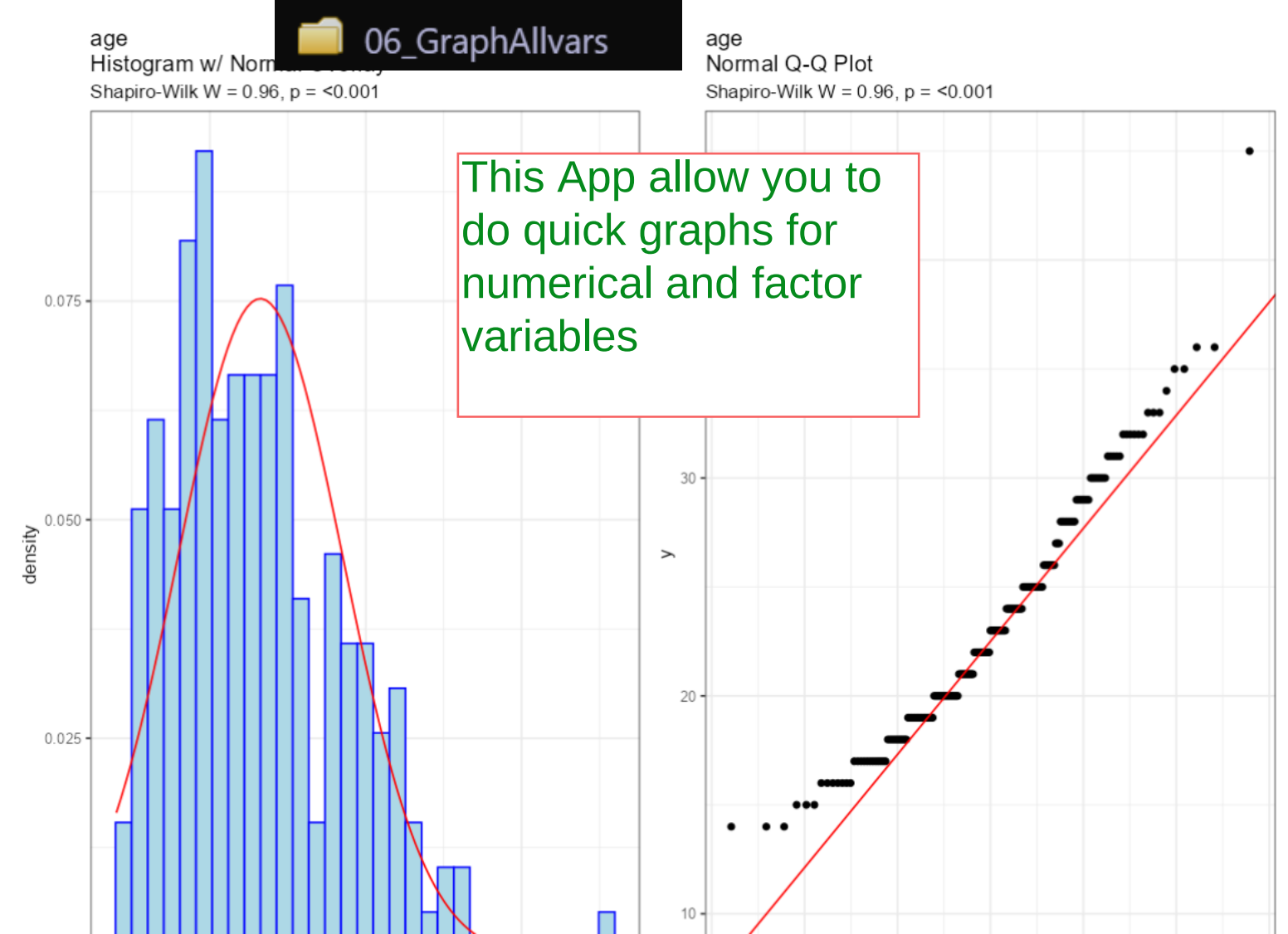
☐ Count
☒ Percent

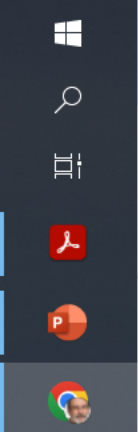
Generate Plots

Word file name (without extension)

All_Plots

Combined Plots





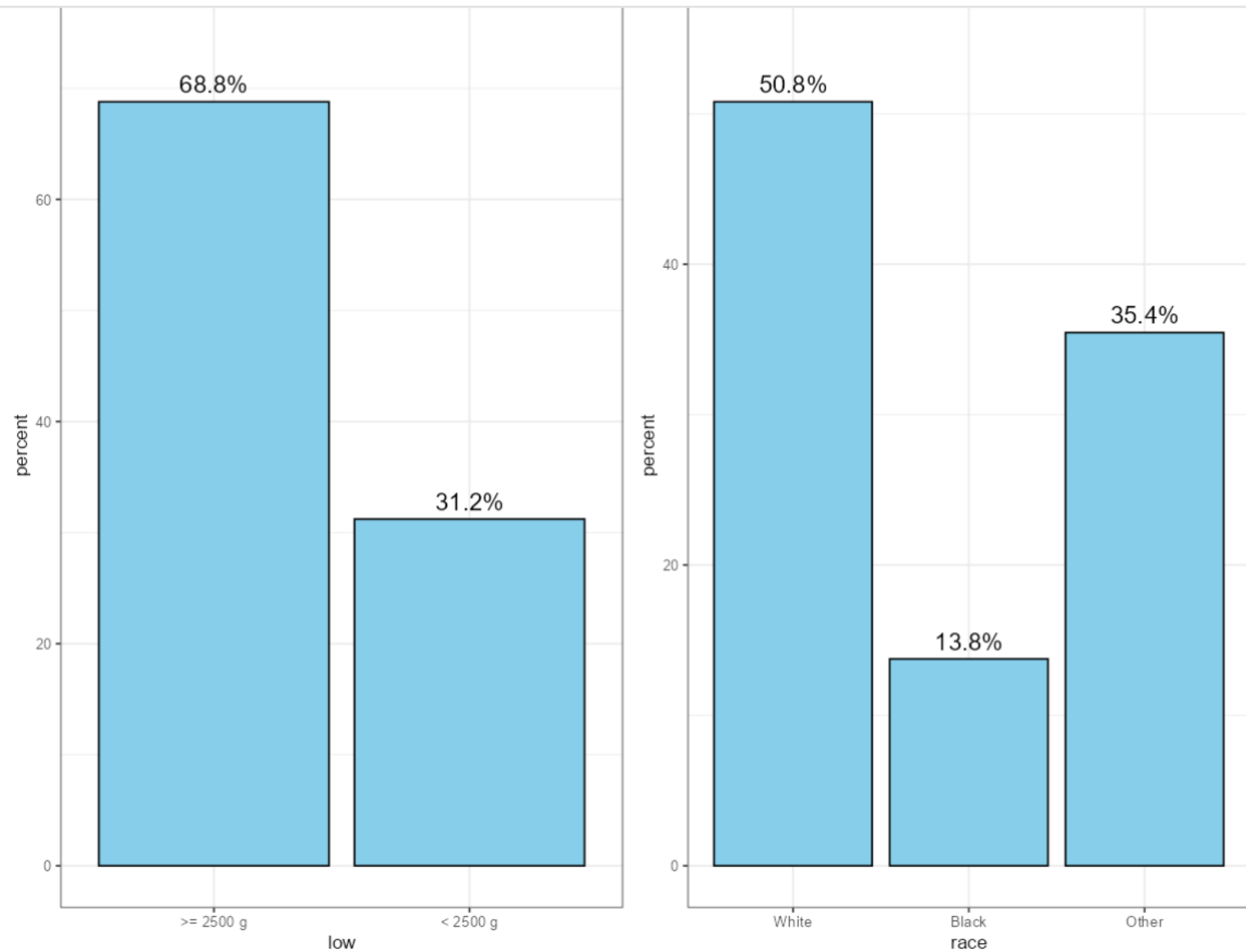
Desktop

>

ENG

18:07
Friday
2026/03/13

3



Unified Graph App: Explore Variables

Data source:
☒ Upload file ☐ R Environment

Upload Excel / Stata / RData / CSV
Browse... lowbwt.RData
Upload complete

07_GraphAllVarsCombined

Continuous Distributions Categorical Distributions Boxplots by Group Scatterplots (Continuous vs Continuous)

Y (numeric):
lwt

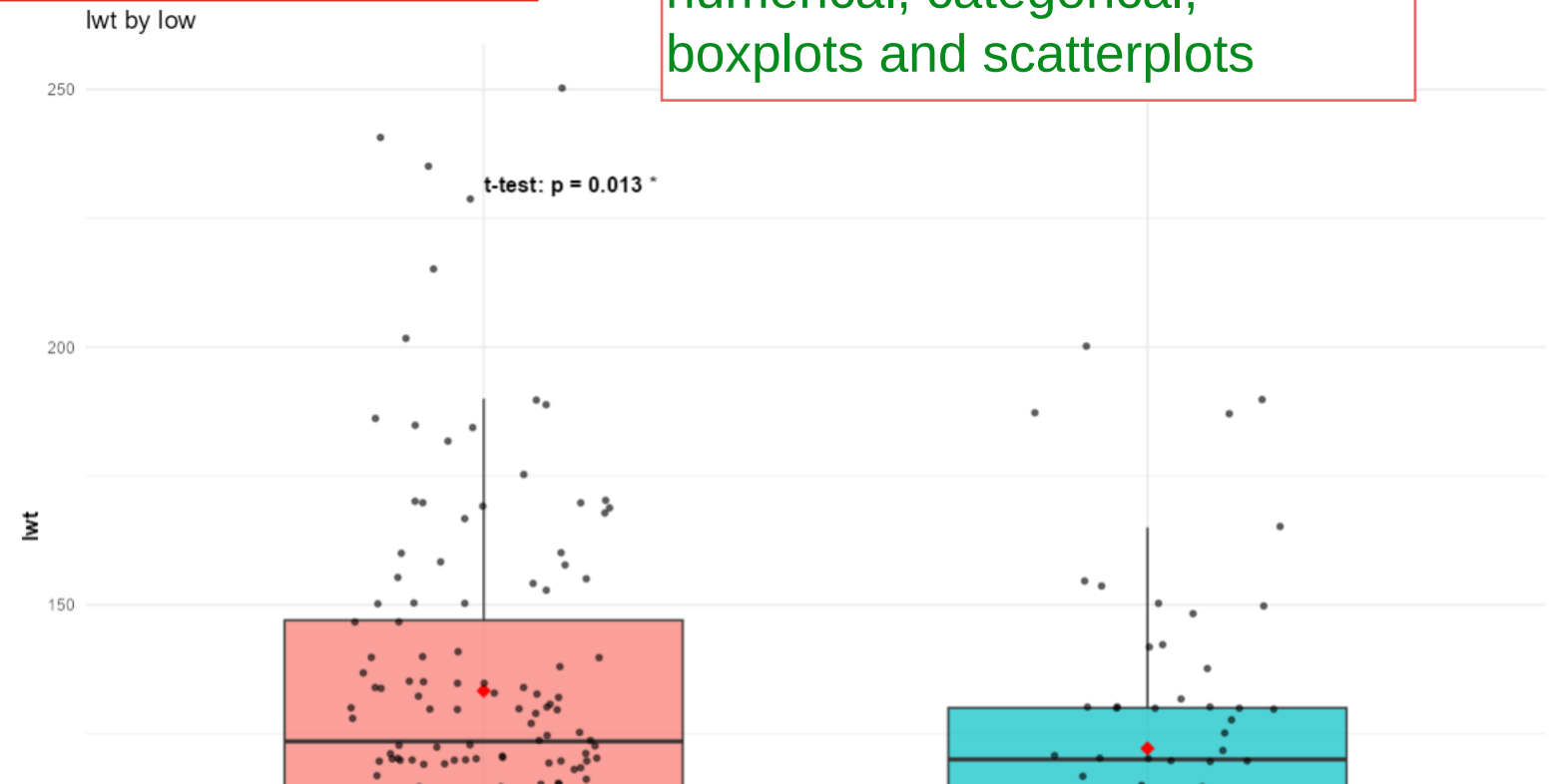
X (group):
low

Facet (optional):
None

Test type:
t-test / ANOVA

P-adjust:
holm

☒ Jitter points
☐ Add violin
☐ Flip axes
☐ Log-10 Y



The App allow you to choose numerical, categorical, boxplots and scatterplots

Unified Graph App: Explore Variables

Data source: ☒ Upload file ☐ R Environment

Upload Excel / Stata / RData / CSV

Browse... lowbwt.RData

Upload complete

Continuous Distributions Categorical Distributions **Boxplots by Group** Scatterplots (Continuous vs Continuous)

Y (numeric):
lwt

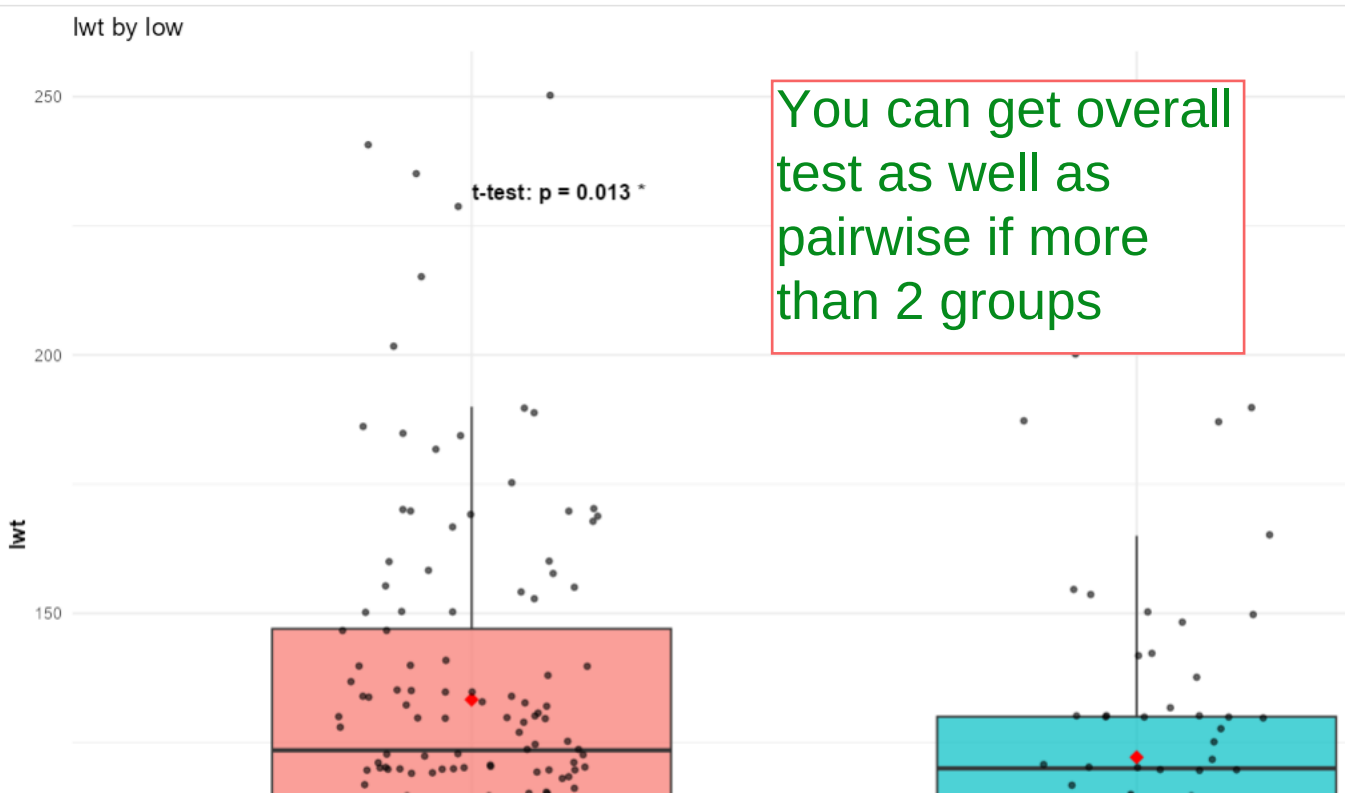
X (group):
low

Facet (optional):
None

Test type:
t-test / ANOVA

P-adjust:
holm

☒ Jitter points
☐ Add violin
☐ Flip axes
☐ Log-10 Y



You can get overall test as well as pairwise if more than 2 groups

Windows taskbar icons: File Explorer, Search, Task View, Adobe Reader, PowerPoint, Chrome, and system tray icons for Desktop, Network, Volume, and Date/Time (18:10 Friday 2026/03/13).

Continuous Distributions Categorical Distributions Boxplots by Group **Scatterplots (Continuous vs Continuous)**

Select variables

X variable:
age

Y variable:
lwt

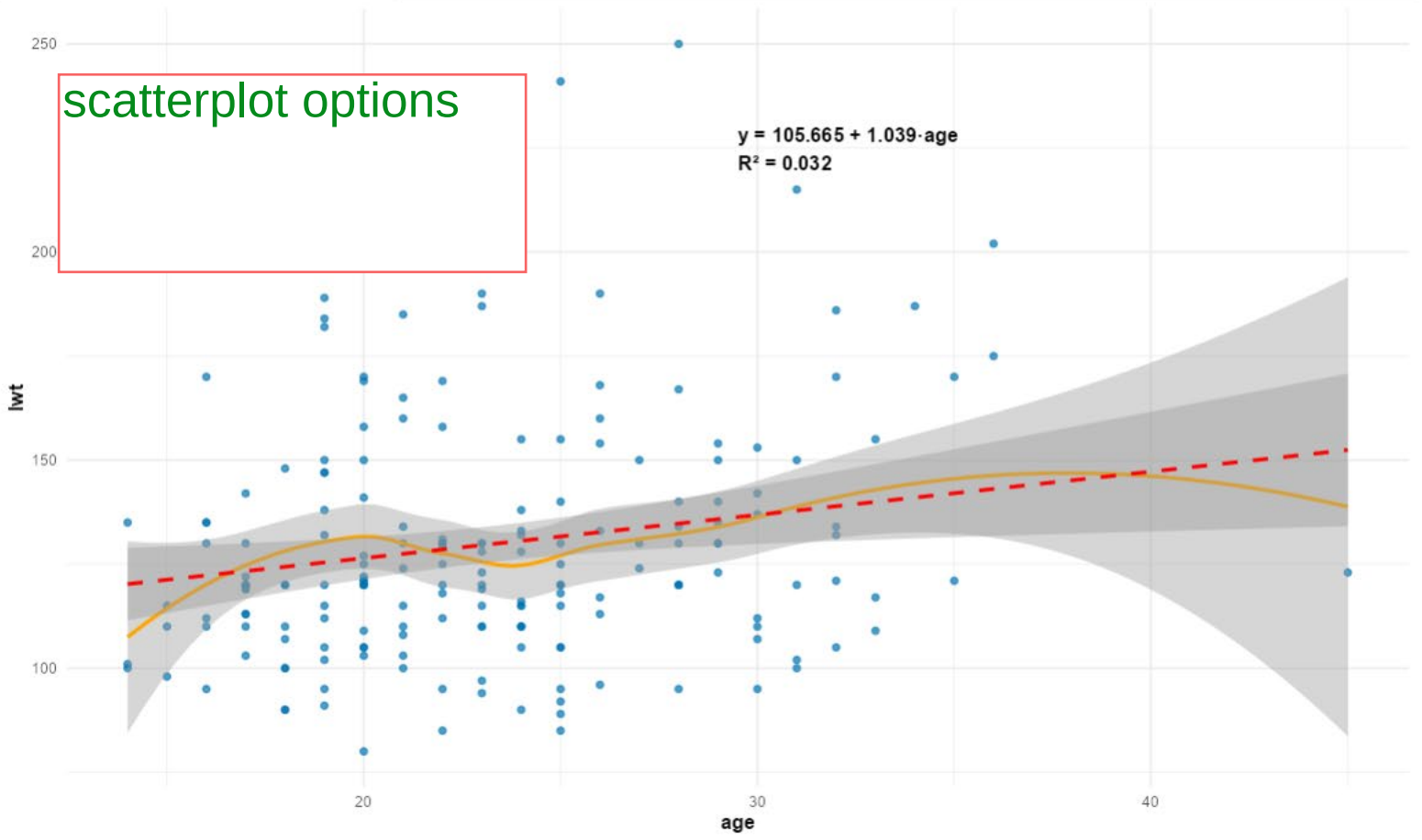
Label variable (optional):
None

☒ Add LOESS smoother
☒ Add linear fit
☒ Show confidence interval
☐ Show row labels

Correlation display
☒ Show correlation on plot

X position (data units):

Y position (data units):



Boxplots with Overall and Pairwise p-values

Data source
☐ Use example: ToothGrowth ☐ Upload CSV/Excel/Stata
☒ Upload RData ☐ Select from environment

Upload RData (.RData/.rda)
Browse... lowbwt.RData
Upload complete

Numeric response (Y)
age

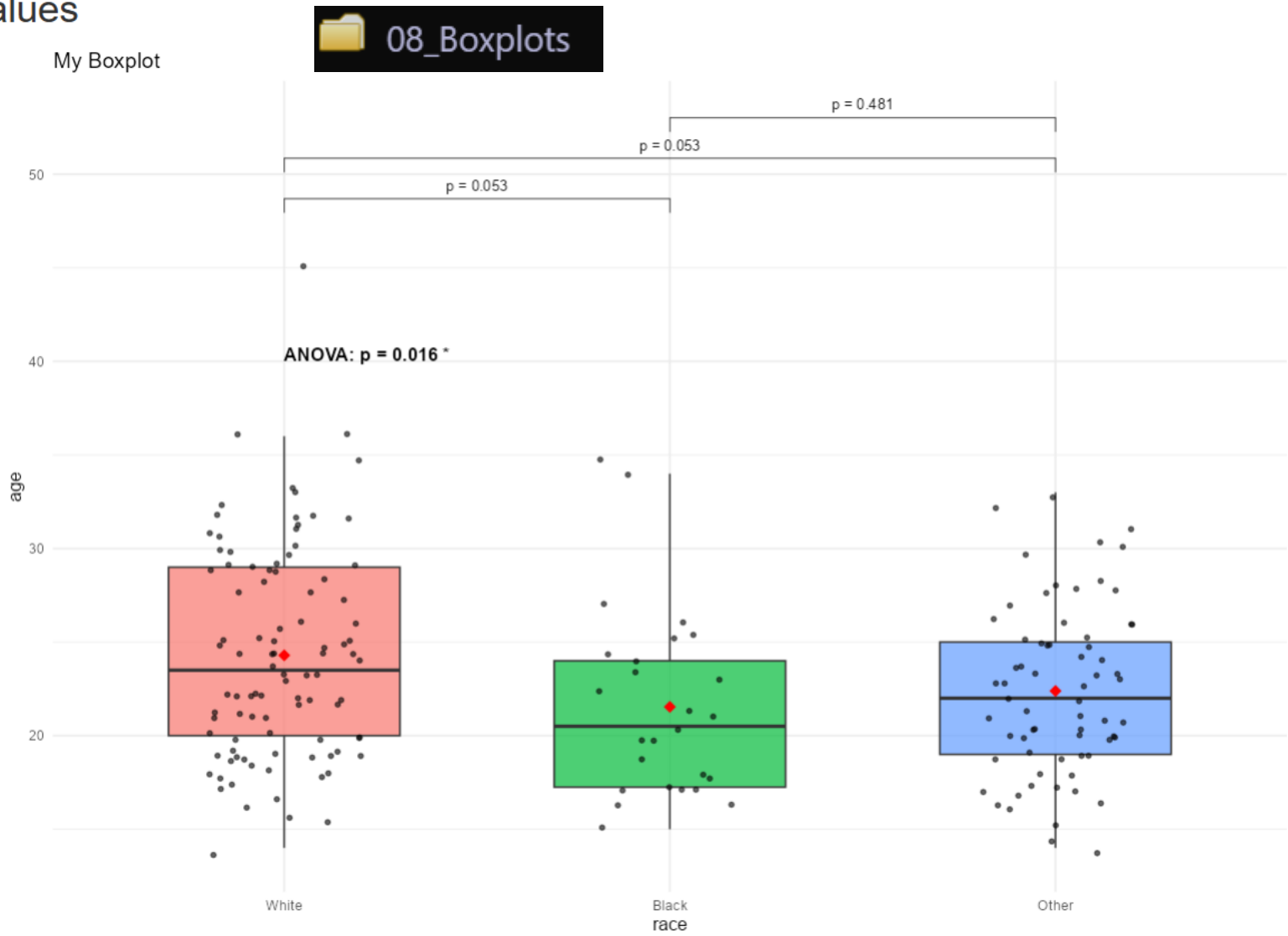
Grouping (X; categorical)
race

Facet (optional)
None

Test type
t-test / ANOVA

P-adjust:
holm

☒ Jitter points
☐ Add violin
☐ Flip axes



 10_Regression

id

In the regression App you can choose linear or logistic regression

	id	low	age	lwt	race	smoke	ptl	ht	ui	ftv	bwt
1	4	< 2500g	28	120	Other	Yes	One	No	Yes	None	709
2	10	< 2500g	29	130	White	No	None	No	Yes	Two, etc.	1021
3	11	< 2500g	34	187	Black	Yes	None	Yes	No	None	1135
4	13	< 2500g	25	105	Other	No	One	Yes	No	None	1330

Load Dataset

Upload Excel, CSV, STATA or RData

Browse...

lowbwt.RData

Upload complete

Or select dataset from R environment:

None

Load Dataset

Model type:

☒ Linear

☐ Logistic

Response:

	Estimate	Std. Error	t value
(Intercept)	2841.961	321.334	8.844
age	-2.036	9.817	-0.207
lwt	3.999	1.737	2.302
raceBlack	-511.535	157.028	-3.258
raceOther	-400.607	119.542	-3.351

Windows taskbar icons: File Explorer, Search, Task View, Edge, PowerPoint, Chrome, Desktop, Network, Sound, ENG, 18:14 Friday 2026/03/13, 3 notifications.

Load Dataset

Upload Excel, CSV, STATA or RData

lowbwt.RData

Upload complete

Or select dataset from R environment:

None

Load Dataset

Model type:

☒ Linear
☐ Logistic

Response:

bwt

Predictors:

age lwt race smoke

☐ Mean-center numeric predictors before interactions

Specify interaction terms (e.g., x1:x2 + x3:x4)

Fit Model



Load Dataset

Upload Excel, CSV, STATA or RData

Browse...

lowbwt.RData

Upload complete

Or select dataset from R environment:

None

Load Dataset

Model type:

☒ Linear

☐ Logistic

Response:

bwt

Predictors:

age lwt race smoke

☐ Mean-center numeric predictors before interactions

Specify interaction terms (e.g., x1:x2 + x3:x4)

Fit Model

Download Outputs

Data PreviewModel SummaryModel CommentaryDiagnostics ContinuousDiagnostics CategoricalPredicted Probabilities

Regression TableVariable ImportanceModel PerformanceCalibration & GOFROC CurveInteraction PlotCoefficient Plot

Coefficient Table (ggstats)

Select Variables

Categorical Variable:

race

Continuous Variable:

lwt

Plots

lwt

White

Black

Other

Sample Quantiles

Theoretical Quantiles

Load Dataset

Upload Excel, CSV, STATA or RData

lowbwt.RData

Upload complete

Or select dataset from R environment:

None

Load Dataset

Model type:

☒ Linear

☐ Logistic

Response:

bwt

Predictors:

age lwt race smoke

☐ Mean-center numeric predictors before interactions

Specify interaction terms (e.g., x1:x2 + x3:x4)

Fit Model

- Data Preview
 - Model Summary
 - Model Commentary
 - Diagnostics Continuous
 - Diagnostics Categorical
 - Predicted Probabilities
 - Regression Table**
 - Variable Importance
 - Model Performance
 - Calibration & GOF
 - ROC Curve
 - Interaction Plot
 - Coefficient Plot
- Coefficient Table (ggstats)

Regression Table (with Global P-values for factors >2 levels)

Regression Table for Outcome: bwt			
Characteristic	Beta	95% CI	p-value
age	-2.0	-21, 17	0.8
lwt	4.0	0.57, 7.4	0.022
race			<0.001
White	—	—	
Black	-512	-821, -202	
Other	-401	-636, -165	
smoke			<0.001
No	—	—	
Yes	-400	-616, -185	

Abbreviation: CI = Confidence Interval

Regression Dashboard

Load Dataset

Upload Excel, CSV, STATA or RData

lowbwt.RData

Upload complete

Or select dataset from R environment:

None

Load Dataset

Model type:

☐ Linear

☒ Logistic

Response:

low

Predictors:

age lwt race smoke

☐ Mean-center numeric predictors before interactions

Specify interaction terms (e.g., x1:x2 + x3:x4)

Fit Model

- [Data Preview](#) [Model Summary](#) [Model Commentary](#) [Diagnostics Continuous](#) [Diagnostics Categorical](#) [Predicted Probabilities](#)
- [Regression Table](#) [Variable Importance](#) [Model Performance](#) [Calibration & GOF](#) [ROC Curve](#) [Interaction Plot](#) [Coefficient Plot](#)

Coefficient Table (ggstats)

Hosmer and Lemeshow goodness of fit (GOF) test

data: actual, fitted(model_fit())
X-squared = 10.842, df = 8, p-value = 0.2108

Brier Score: 0.1952

Interpretation of the Brier Score:

- **Definition:** The Brier Score measures the mean squared difference between predicted probabilities and the actual 0/1 outcomes.
- **Range:** 0 to 1. Lower is better — 0 indicates perfect probabilistic predictions.
- **Typical Values:** Around 0.25 for random guessing (if outcome prevalence ≈ 0.5).
- **Meaning:** It combines calibration (how close predicted probabilities are to true probabilities) and discrimination (how well predictions separate cases from non-cases).
- **In practice:** Smaller values indicate more accurate, better-calibrated models.

Summary: The Brier Score is like an RMSE for predicted probabilities — lower means better overall probability accuracy.

Basic Calibration Plot



Response:

low

Predictors:

age lwt race smoke

☐ Mean-center numeric predictors before interactions

Specify interaction terms (e.g., $x_1:x_2 + x_3:x_4$)

Fit Model

Download Outputs

File name prefix:

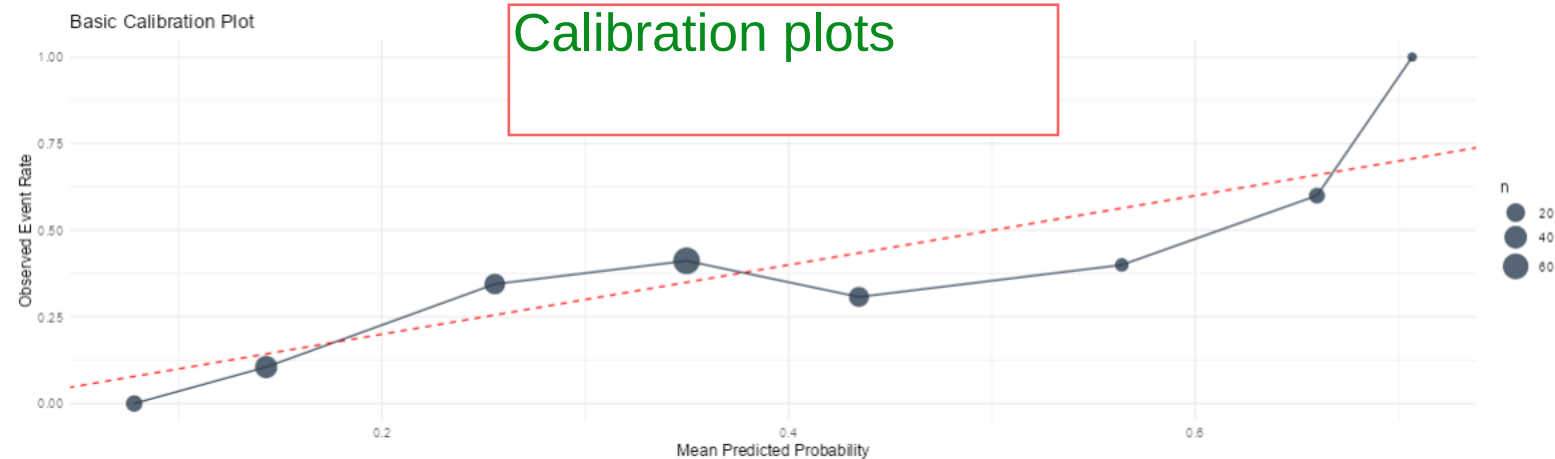
regression

[Download Word Report](#) [Download Plots ZIP](#)

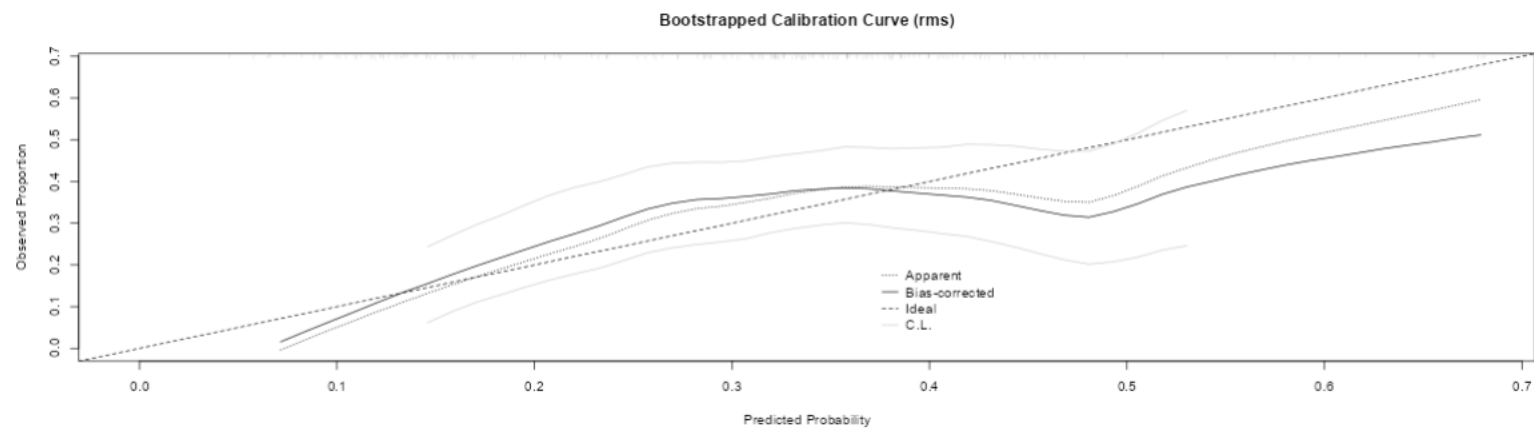
- **typical values:** Around 0.25 for random guessing (if outcome prevalence = 0.5).
- **Meaning:** It combines calibration (how close predicted probabilities are to true probabilities) and discrimination (how well predictions separate cases from non-cases).
- **In practice:** Smaller values indicate more accurate, better-calibrated models.

Summary: The Brier Score is like an RMSE for predicted probabilities — lower means better overall probability accuracy.

Basic Calibration Plot



Bootstrapped Calibration Plot (rms)



Windows taskbar showing icons for File Explorer, Edge, PowerPoint, and a user profile picture.

Regression Dashboard

Load Dataset

Upload Excel, CSV, STATA or RData

Browse...

lowbwt.RData

Upload complete

Or select dataset from R environment:

None

Load Dataset

Model type:

☐ Linear

☒ Logistic

Response:

low

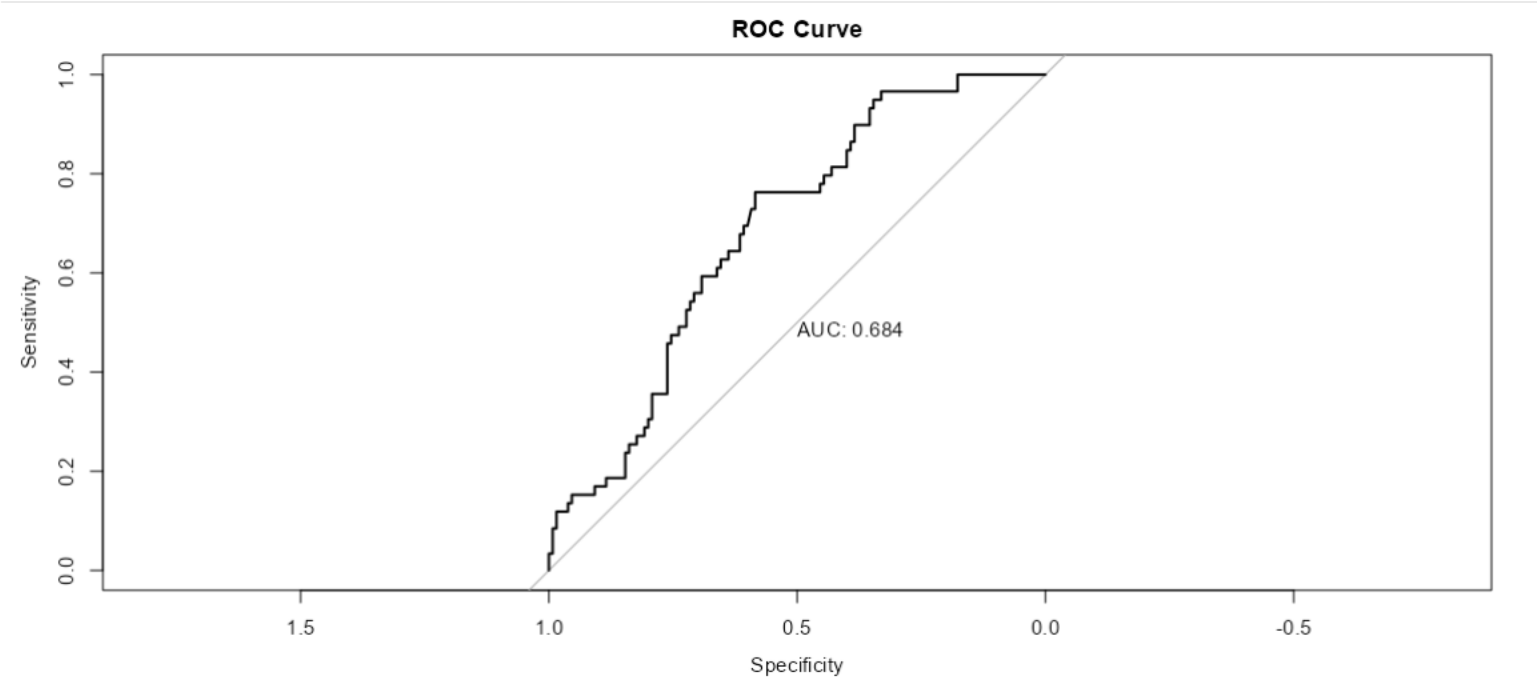
Predictors:

age lwt race smoke

☐ Mean-center numeric predictors before interactions

Specify interaction terms (e.g., x1:x2 + x3:x4)

- Data Preview
- Model Summary
- Model Commentary
- Diagnostics Continuous
- Diagnostics Categorical
- Predicted Probabilities
- Regression Table
- Variable Importance
- Model Performance
- Calibration & GOF
- ROC Curve
- Interaction Plot
- Coefficient Plot
- Coefficient Table (ggstats)



Upload Excel, CSV, STATA or RData

Browse... lowbwt.RData
Upload complete

Or select dataset from R environment:

None
Load Dataset

Model type:

☐ Linear
☒ Logistic

Response:

low

Predictors:

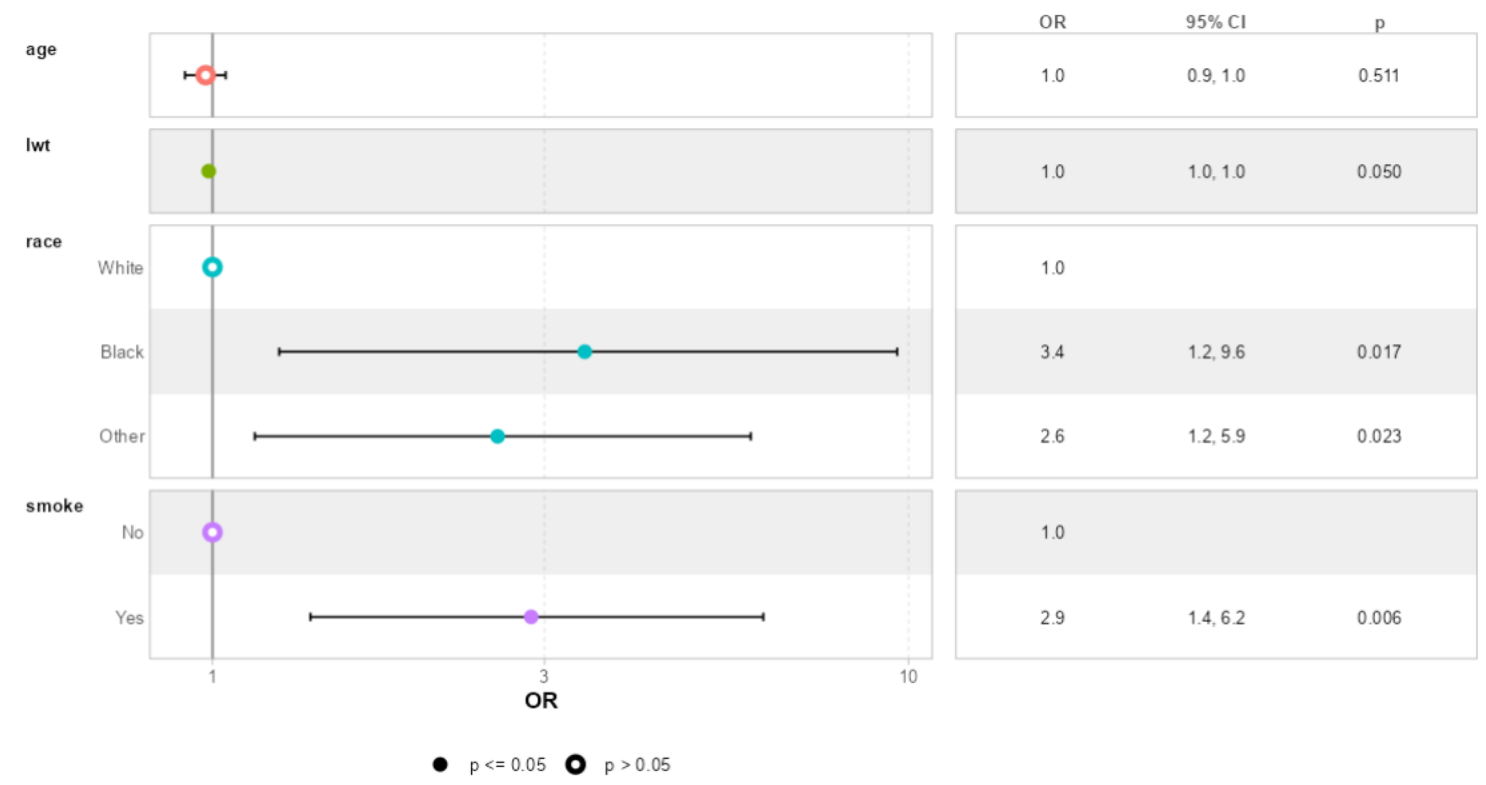
age lwt race smoke

☐ Mean-center numeric predictors before interactions

Specify interaction terms (e.g., x1:x2 + x3:x4)

- Diagnostics Categorical
- Predicted Probabilities
- Regression Table
- Variable Importance
- Model Performance
- Calibration & GOF
- ROC Curve
- Interaction Plot
- Coefficient Plot

Coefficient Table (ggstats)



<https://github.com/paulrheeder-rgb/shinyappsforbiostats>

Demo of Apps



paul.rheeder@up.ac.za

-  01_DataDictionary
-  02_PostImport
-  03_DplyrDataEdit
-  04_Table1AllStats
-  05_Table1or2
-  06_GraphAllvars
-  07_GraphAllVarsCombined
-  08_Boxplots
-  09_Scatterplots
-  10_Regression
-  11_Missing
-  12_ImputedAnalysis
-  13_RMSSplines
-  14_MixedModelLinear

That was the introduction. Now get these and use them!